Quantitative Financial Risk Management

Duisenberg Honours Programme in Quantitative Risk Management

Period 4.5

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Part I

Challenges & Basics

(McNeil, Frey, and Embrechts, 2015, Ch. 1-2, Part I)

1. Motivation
2. Foundations
3. Risk Measures: VaR and ES
4. Review

What Risk? Why Risk?

(See Simon Sinek, TED 2016)

What is Risk?

- "hazard, a chance of bad consequences, loss or exposure to mischance" (Oxford English Dictionary)
- "any event or action that may adversely affect an organization's ability to achieve its objectives and execute its strategies"
- "the quantifiable likelihood of loss or less-than-expected returns"

(MFE §1.1)

This course

- Assumes good knowledge of econometrics+analysis+statistics+linear algebra+programming
- Revisits risk theory
- Moves on through measurement and modelling
- To allow you to work in the field...

Types of financial risks

- Market risk
- Credit risk
- Liquidity risk
- Model risk
- Operational risk
  plus e.g.
- Business risk
- Underwriting risk
- Cyber risk
- Legal risk

etc...

Where do the major risks come from?

- Market risk: risk of a change in the value of a financial position due to changes in the value of the underlying components on which that position depends, such as stock and bond prices, exchange rates, commodity prices
- Credit risk: ...
- Liquidity risk: ...
- Model risk: ...
- Operational risk: ...

(MFE §1.1.2)

Where do the major risks come from?

- Market risk: ...
- Credit risk: risk of not receiving promised repayments on outstanding investments such as loans and bonds, because of the "default" of the borrower
- Liquidity risk: ...
- Model risk: ...
- Operational risk: ...

(MFE §1.1.2)

Where do the major risks come from?

- Market risk: ...
- Credit risk: ...
- Liquidity risk: risk of not being able to roll over debt, or risk associated with not being able to close existing positions due to lack of liquidity on markets
- Model risk: ...
- Operational risk: ...

(MFE §1.1.2)

Where do the major risks come from?

- Market risk: ...
- Credit risk: ...
- Liquidity risk: ...
- Model risk: risk associated with models used for pricing, risk management etc being "wrong": inappropriate model, wrong parameters, wrong calibration, wrong use, ...
- Operational risk: ...

(MFE §1.1.2)

Where do the major risks come from?

- Market risk: ...
- Credit risk: ...
- Liquidity risk: ...
- Model risk: ...
- Operational risk: risk of losses resulting from inadequate or failed internal processes, people and systems, or external events

(MFE §1.1.2)

Where do the other risks come from?

- ...
- Business risk: Changing tastes, insufficient profits, government policy changes
- Underwriting risk: Insurances leading to larger payouts than inlays, problems spreading underwriting of large risks
- Cyber risk: Ransomware attacks, hacking
- Legal risk: Unforeseen lawsuits, possibly due to unlawful behaviour of subsidiaries or employees

Market Risk: (Generally) unfavorable changes due to movements in

- Stocks and stock indices
- Exchange rates
- Interest rates
- Other asset prices: Commodities, real estate funds etc

What do we need? Challenges?

- Mathematical rigor and precision in current practices
- Going beyond current practices

This means we're dealing with

- Dependence of risks and multivariate nature of problem
- Extreme values and heavy tails
- Concentration risk
- Scale of portfolios and liquidity risk
- Model risk
- Data issues
- Understanding current/future regulation

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Focus for the course:

Aggregate risk

- ... depending on multivariate, high dimensional vectors
- ... of underlying risk factors
- ... which may all depend on each other,
- ... and more so in case of extreme events...

Who says so?

> *Extreme, synchronized rises and falls in financial markets occur infrequently but they do occur. The problem with the models is that they did not assign a high enough chance of occurrence to the scenario in which many things go wrong at the same time — the "perfect storm" scenario.*

(Business Week (9/98), MFE §1.5.2)

Who says so?

> *From the point of view of the risk manager, inappropriate use of the normal distribution can lead to an understatement of risk, which must be balanced against the significant advantage of simplification. From the central bank's corner, the consequences are even more serious because we often need to concentrate on the left tail of the distribution in formulating lender-of-last-resort policies. Improving the characterization of the distribution of extreme values is of paramount importance.*

(Alan Greenspan, 1995)

Who says so?

> *With globalisation increasing, you'll see more crises. Our whole focus is on the extremes now - what's the worst that can happen to you in any situation - because we never want to go through that (=LTCM) again*

(John Meriwether, The Wall Street Journal, 21/8/2000)

Diversification is meant to lower risks. But what if, in a perfect storm, all correlations increase to 1?
Then **there is no diversification left**!
Practitioners describe this also as a concentration of risk.

> *Over the last number of years, regulators have encouraged financial entities to use portfolio theory to produce dynamic measures of risk. VaR, the product of portfolio theory, is used for short-run, day-to-day profit and loss exposures. Now is the time to encourage the BIS and other regulatory bodies to support studies on stress test and concentration methodologies. Planning for crises is more important than VaR analysis. And such new methodologies are the correct response to recent crises in the financial industry.*

(M. Scholes, 2000)

A further challenge in QRM is the typical scale of the portfolios under consideration; in the most general case, a portfolio may represent the entire position in risky assets of a financial institution.
Calibration of detailed multivariate models for all risk factors is an almost impossible task, and any sensible strategy must involve dimension reduction; that is to say, the identification of key risk drivers and a concentration on modelling the main features of the overall risk landscape.

(MFE §1.5.2)
⇒ We'll need a broad brush approach...

Quantitative Financial Risk Manager of the future should have a combined skill set that includes concepts, techniques and tools from many fields:

- Mathematical Finance
- Statistics and Financial Econometrics
- Actuarial Mathematics
- Non-quantitative skills, especially *communication* skills
- Humility — QFRM is a small piece of a bigger picture!

How do you fit in?

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How do you fit in?

- FRTB, review of the trading book
- TRIM, trade related investment measures
- IRRBB, interest rate risk in banking book
- IFRS 9, 17, International Financial Reporting Standards
- Basel III, International Regulatory Standards for Banks
- Solvency II, EU Directive on insurance regulation
- IORP, on social security and retirement benefits
- GDPR, privacy
- Data Act and Data Governance Act, access to and safe sharing of data
- ...

(partial list...)

Considered known:

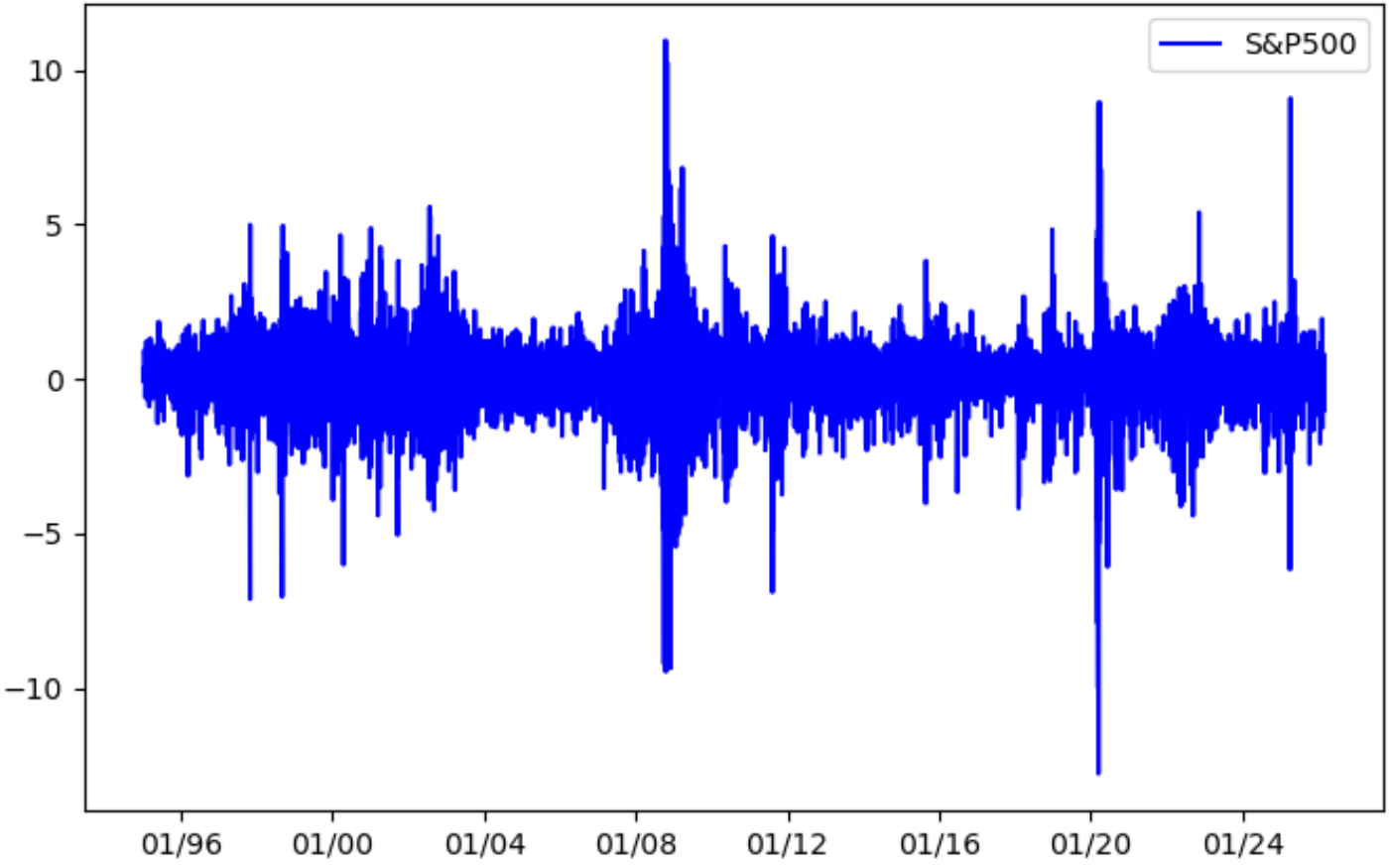
- Volatility
- Portfolio variance and Volatility
- VaR (Value-at-Risk) and ES (Expected Shortfall)
- Basic Methods: Historical, Variance/covariance, $\mathcal{N}$ and t
- EWMA, (G)ARCH

Considered new:

- Filtered historical simulation
- Backtesting of RM models

- Financial returns: random variables
- Horizon: day, week, month, year $\rightarrow$ depends on financial institution/its department, risk type, etc
- We model random variables by their probability distribution (full model) or their moments
- First moment: average, or expected return
- Second (central) moment: variance (squared volatility)
- Third (central) moment: related to skewness
- Fourth (central) moment: related to kurtosis

Ex: S&P 500



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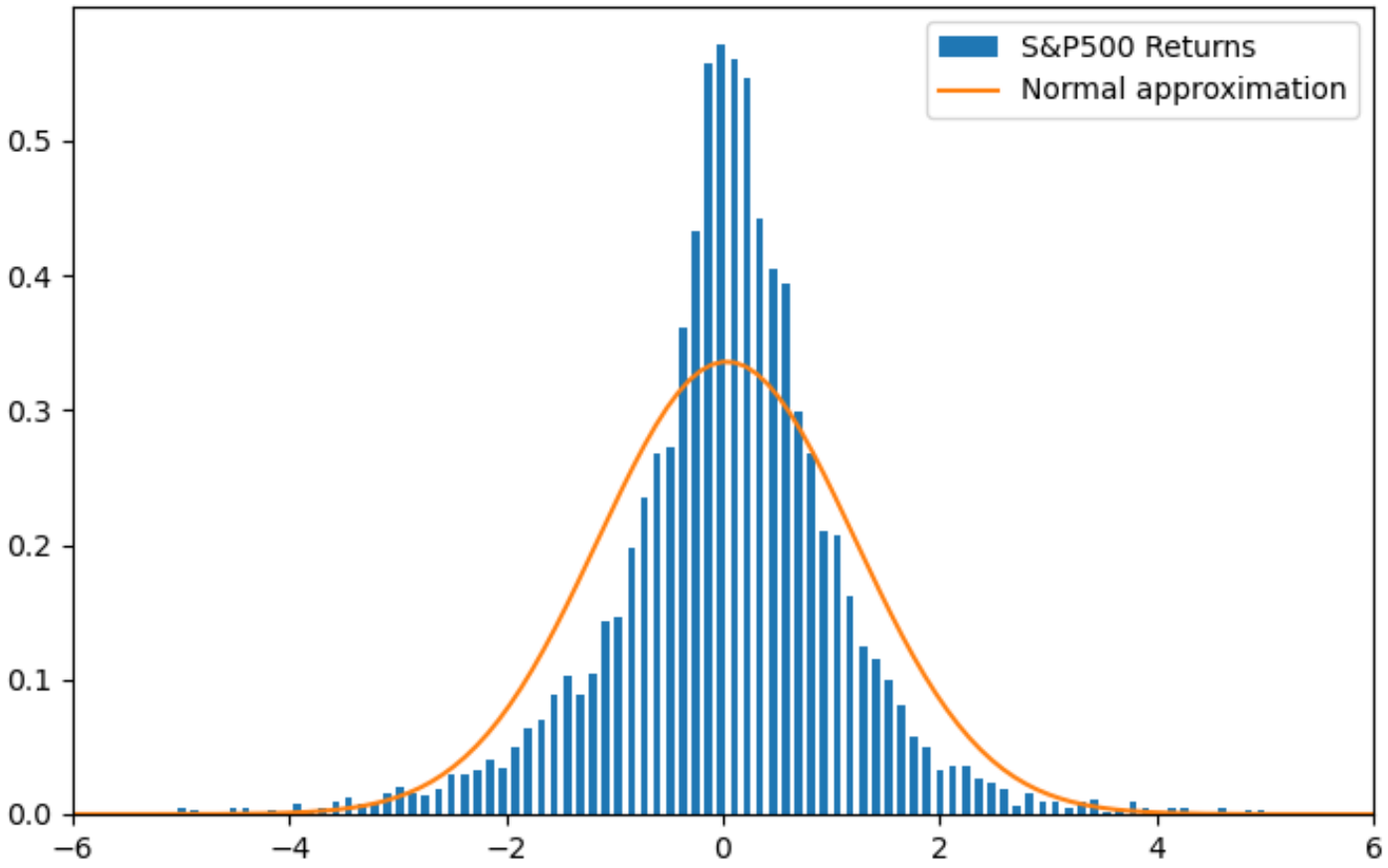
S&P500 Returns
Normal approximation
0.5
0.4
0.3
0.2
0.1
0.0
−6
−4
−2
0
2
4
6

For getting variance/volatility:

- Define σ_t as *volatility*, σ_t^2 the *variance* on day t
- Calculate e.g. *historical* variance over m days, $\sigma_t^2 \overset{\text{hist}}{=} \frac{1}{m}\sum_{i=1}^{m} r_{t-i}^2$
- Use (adjusted?) *close* asset prices $S_t, t = T - m + 1, \ldots, T$,
- with *continuously compounded* r_t or *simple* returns R_t
- take $\sqrt{\ }$ to get volatility, $\sigma_t \equiv \sqrt{\sigma_t^2}$

	Cont. compounded	Simple
Return	$r_t = 100 \ln \frac{S_t}{S_{t-1}}$	$R_t = 100 \frac{S_t - S_{t-1}}{S_{t-1}}$
Mean	$\bar{r}_t = \frac{1}{m}\sum_{i=1}^{m} r_{t-i}$	$\bar{r}_t \equiv 0$
Variance	$\sigma_t^2 = \frac{1}{m-1}\sum_{i=1}^{m}(r_{t-i} - \bar{r}_t)^2$	$\sigma_t^2 = \frac{1}{m}\sum_{i=1}^{m} r_{t-i}^2$

(Scaling to % — Assuming mean zero — Divide by m or $m-1$ — Cont. compounded vs simple. See Brooks (2014, §1.4))

- Volatility usually measured as % per year
- In RM/VaR, we need daily volatility

$$\sigma_{\text{day}} = \frac{1}{\sqrt{250}} \sigma_{\text{year}}$$

- Similar square root of time scaling rule can be used to get weekly or monthly volatility

Q: Why 250? What other values are used?

Expected return and variance of N-assets portfolio

If we have N assets with returns $\boldsymbol{r}_t = (r_{1t}, \ldots, r_{Nt})$, in a portfolio with weights $\boldsymbol{w} = (w_1, \ldots, w_N)$, with $\sum w_i = 1$.
Then the portfolio delivers

$$E(r_p) = \sum_{i=1}^{N} w_i E(r_{it}) = \boldsymbol{w}' E(\boldsymbol{r}_t) \tag{1.1}$$

$$\mathrm{Var}(r_p) = \sum_{i=1}^{N} w_i^2 \mathrm{Var}(r_i) + 2 \sum_{i=1}^{N} \sum_{j=i+1}^{n} w_i w_j \mathrm{Cov}(r_i, r_j) = \boldsymbol{w}' \Sigma_r \boldsymbol{w} \tag{1.2}$$

Needed:

- Variance/covariance matrix of returns

Q: Check yourself for $N = 2$, then extend to $N = 3, \ldots$, from standard properties of variance,
$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\mathrm{Cov}(X, Y)$.

Variance of equally weighted N-assets portfolio

If $\mathrm{Cov}(r_i, r_j) = \rho\sigma^2$, $w_i = w_j = \frac{1}{N}$, $\sigma_i = \sigma_j = \sigma \ \forall i \neq j$, then

$$\mathrm{Var}(r_p) = \frac{\sigma^2}{N} + \frac{N-1}{N}\rho\sigma^2 \overset{N\to\infty}{\to} \rho\sigma^2 \equiv \mathrm{Cov}(r_i, r_j)$$

Q: Prove this.

If covariances and variances are different, then

$$\mathrm{Var}(r_p) = \frac{\overline{\sigma}^2}{N} + \frac{N-1}{N}\overline{\mathrm{Cov}(r_i, r_j)}$$

(variance of portfolio relates to average variance of assets and average covariance of assets)

Q: Prove this as well.

What if we have a risk-free security (e.g. cash), investing fraction x in risky portfolio?
Expected return, variance, and volatility:

$$E(r_{xp}) = (1-x)r_f + xE(r_p) = r_f + x[E(r_p) - r_f] \tag{1.3}$$

$$\begin{aligned}\mathrm{Var}(r_{xp}) &= (1-x)^2\mathrm{Var}(r_f) + x^2\mathrm{Var}(r_p) + 2(1-x)x\mathrm{Cov}(r_f, r_p) \\ &\equiv x^2\mathrm{Var}(r_p) \qquad (1.4)\end{aligned}$$

$$\sigma(r_{xp}) = x\sigma(r_p) \tag{1.5}$$

Q: Why does the equation simplify?

Notes on Leverage:

- Borrowing money to invest in risky assets.
- Or shorting some risky assets to invest in other risky assets.
- Such portfolios are called leveraged portfolios.

Bottomline: These are more risky than unleveraged positions.

Ex 1: Leverage
You have 1000 euros and you borrow another 1000 (against interest of e.g. 5%). You invest all 2000 in a stock (or stock index) with volatility of 20% and expected return of 10%.
$\Rightarrow$ Your resulting portfolio has expected return of $E(r_p) = 15\%$= $2 \times 10\%$-5%, but the volatility of $\sigma_p = 2 \times 20\%$=40%!
Q: Write this out as a portfolio investment into a risk-free and a risky investment.

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Problem (Expected return and volatility with a short sale)

Given: Suppose you have \$20k in cash to invest. You decide to short sell \$10k worth of Coca-Cola stock and invest the proceeds from your short sale, plus your \$20k, in Intel. Volatilities are $\sigma_c = 0.25, \sigma_i = 0.5$, *and correlation is zero (of course it is not, but use this as a lower bound). Expected returns* $E(r_c) = 6\%, E(r_i) = 26\%$

Q: *What is the expected return and volatility of your portfolio?*
Q2: *What changes if correlation increases to* $\rho = 0.2$*?*

Investment= Investment in CC + Investment in I= -10K + 30K= 20K.
Hence the weights become

$$x_i = \frac{30k}{20k} = 1.5 = 150\%$$
$$x_c = \frac{-10k}{20k} = -0.5 = -50\%$$

Then use the (1.1) and (1.2) to find

$$E(r_p) = x_i Er_i + x_c Er_c = 1.5 \times 26\% - 0.5 \times 6\% = 36\%$$
$$\begin{aligned}\mathrm{Var}(r_p) &= x_i^2 \mathrm{Var} r_i + x_c^2 \mathrm{Var} r_c + 2x_i x_c \mathrm{Cov}(r_i, r_c) \\ &= 1.5^2 \times .5^2 + (-0.5)^2 \times .25^2 + 2(1.5)(-0.5) \times 0 = 0.578\end{aligned}$$

hence

$$\sigma(r_p) = 0.76 = 76\%$$

Or: Short selling increases the expected return of your portfolio, but also the volatility, above those of the individual stocks.

Given time horizon h, confidence level/probability α,

what is the loss L that, with probability α, will not be exceeded, holding a portfolio for h days?

or

$$\begin{aligned}\mathrm{VaR}_\alpha = \mathrm{VaR}_\alpha(L) &= \inf\{l \in \Re : P(L > l) \leq 1 - \alpha\} \\ &= \inf\{l \in \Re : P(L < l) \geq \alpha\} \\ &= q_\alpha(F_L) = F_L^{-1}(\alpha)\end{aligned}$$

MFE Def 2.8

That is: In $1 - \alpha$ of the cases, the loss on the portfolio will be at most VaR_α dollars.

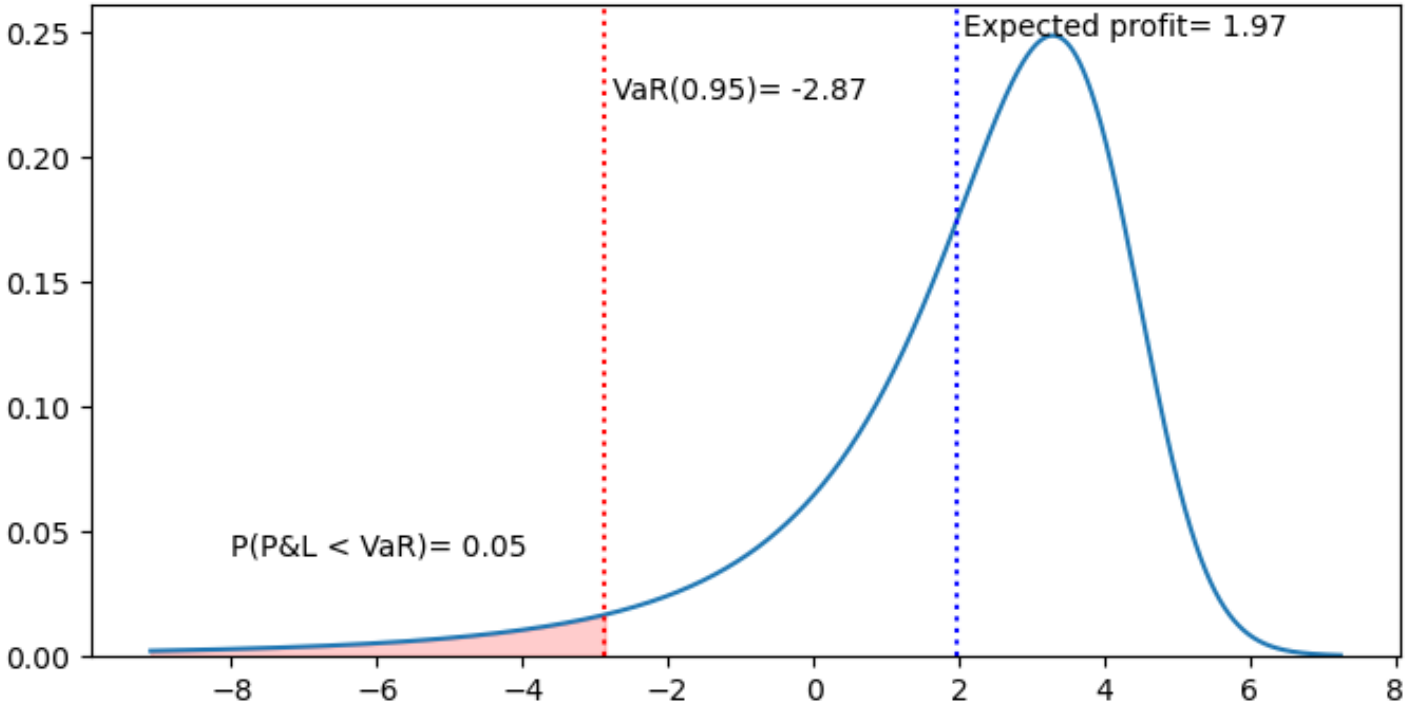
Expected shortfall is the average loss when VaR is exceeded:

$$\mathrm{ES}_\alpha = E(L|L > \mathrm{VaR}_\alpha)$$

So, given the loss exceeded VaR, how much money I will lose on average?

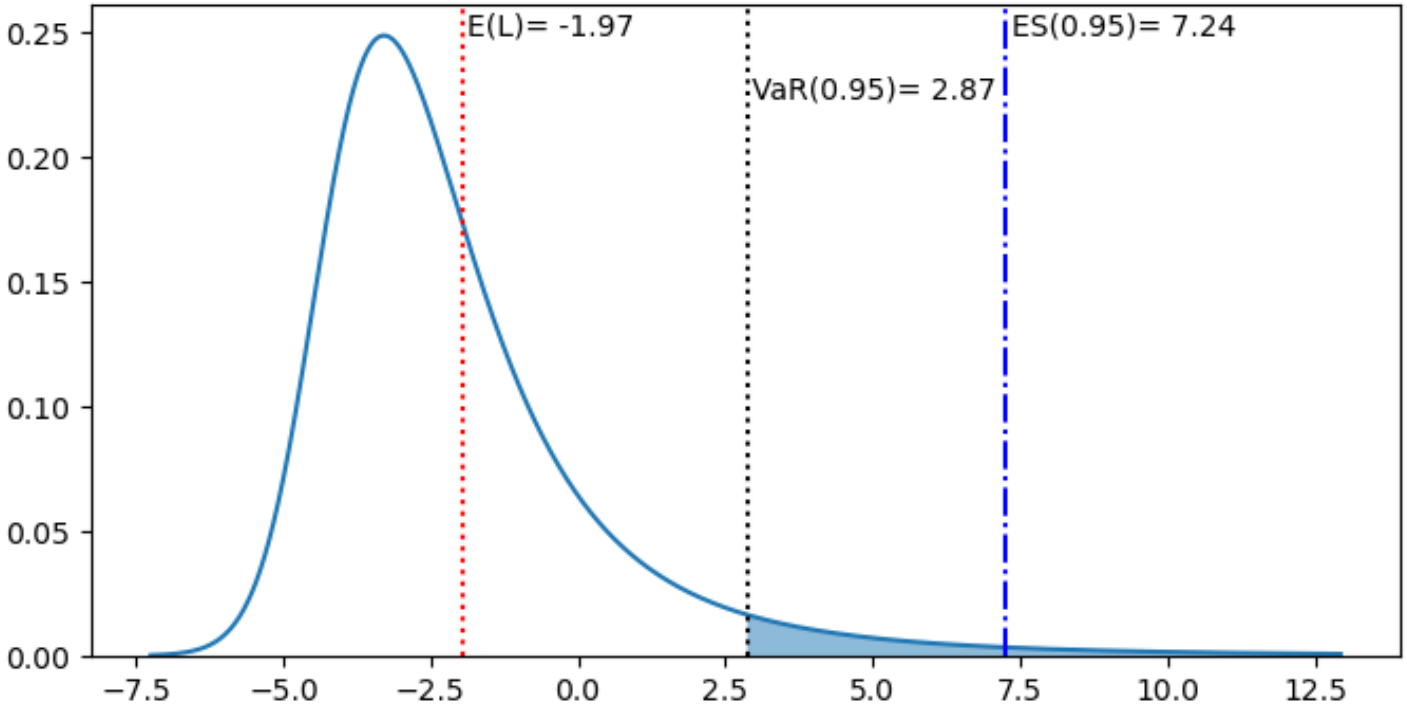
- ES gives information about size of large losses, once they occur, while VaR gives information about frequency of large losses but says little about potential size.

VaR in visual terms



Example of Profit & Loss distribution, with 95% VaR

VaR in visual terms



Example of Loss distribution, with 95% VaR and 95% ES

- In QRM course (and in market risk management in general), we always work with LOSS distribution and not P/L distribution
- So losses are positive, and gains are negative
- This is because we are focused on losses and at the same time, we like working with positive random variables

- It captures an important aspect of risk in a single number
- It is easy to understand
- It asks the simple question: "How bad can things get?"
- Disadvantage: it gives you a minimum loss which occurs with a certain probability
- $\Rightarrow$ It does not tell you how bad it could get, once you cross this threshold.

Expected Shortfall (ES) overcomes this disadvantage.

VaR vs. ES

- VaR is the loss level that will not be exceeded with a specified probability
- Expected Shortfall is the expected loss given that the loss is greater than the VaR level
- ES is theoretically more appealing. It also contains more information about large losses than VaR.

Problem with VaR (more about this later)

VaR is not a "coherent" risk measure, i.e. does not always reflect benefits of diversification:

$$\mathrm{VaR}(P1 + P2) >\overset{?}{=}< \mathrm{VaR}(P1) + \mathrm{VaR}(P2)$$

Expected Shortfall however is better in this respect.

- Regulators used to base the capital they require banks to keep on VaR
- Pre-FRTB: The market-risk capital was $C = k \times \mathrm{VaR}(h = 10, q = 99\%)$, where $k \geq 3$
- Post-FRTB: in 2015, BIS changed its capital requirements calculations. Now it is $C = k \times \mathrm{ES}(h = 10, q = 97.5\%)$, for some $k \geq 3$.

Q: Why change in confidence level?

What have we done, from McNeil, Frey, and Embrechts (2015, MFE):

- Chapter 1
- Chapter 2: 2.1 and 2.3 (until 2.3.5: Coherent and Convex RM (the rest of Chapter 2 will come a bit later))
- Chapter 3

Review: What should you do?

Read/study the above chapters, in relation to the slides, and then

- Refresh your knowledge about concepts discussed today: distributions and moments, quantiles of a distribution, portfolio variance etc.
- Do questions asked in slides
- Work through L2 using the video: Refresh volatility measurement
- Do Exercises 1 t/m 20 in HW Basic (`qfrm_ex_basic.pdf`)
- Start on Assignment 1: form groups, gather data, ...

Brooks, Chris (2014). *Introductory Econometrics for Finance*. 3rd. Cambridge: Cambridge University Press.

McNeil, Alexander J., Rüdiger Frey, and Paul Embrechts (2015). *Quantitative Risk Management: Concepts, Techniques, and Tools*. Revised. Princeton: Princeton University Press. ISBN: 978-0-691-16627-8.